

ODUFISAN, OLUWASEUN ISAAC

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SUMMARY

I am a researcher and full stack software engineer with interests in AI safety, LLMs, generative AI, multimodal AI, human-AI interaction, NLP, and responsible AI, committed to creating impactful, secure, and socially responsible technologies.

EDUCATION

The Federal University of Technology, Akure (FUTA), Akure, Ondo State, Nigeria. (March 2024).

Bachelor of Engineering in Computer Engineering (First Class Hons.) CGPA: 4.79/5.00. (Top 1%)

National Institute of Information Technology (NIIT), Ikeja Centre, Lagos State, Nigeria. (December 2018).

Diploma in Java Technologies

Yaba College of Technology (YABATECH), Yaba, Lagos State, Nigeria. (March 2017).

National Diploma in Computer Engineering (Upper Credit) CGPA: 3.22/4.0

HONOURS AND AWARDS

- Vice Chancellor Merit Award:** First Class Graduate of The Federal University of Technology, Akure (November 2024).
 - HOD Merit Award:** Best Graduating Student of the Department of Computer Engineering (November 2024).
 - Class of Aurelius SEET Award:** The School of Engineering and Engineering Technology (SEET) Aurelius Set Award of Academic Excellence for Being a Top Student in the Department of Computer Engineering (March 2024).
 - Chronicles SDC Staff of the Month:** Award for Commitment to Excellence at Work and Positive Impact (February 2023).
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UNDERGRADUATE THESIS PROJECT

Facial Age Estimation Mobile Application (February 2024).

- Developed a mobile application for facial age estimation targeting Black individuals, utilizing a deep ensemble learning model built with ResNet50 and Keras (with focus on improving accuracy).
 - Integrated React Native for the front-end, Django for the back-end, and SQLite for efficient data management. Integrated React Native for the front-end, Django for the back-end, and SQLite for efficient data management.
 - Conducted extensive data collection across Nigeria and Africa to support model training.
 - Presented the project as an undergraduate thesis at a research seminar.
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PUBLICATION

Oluwaseun Isaac Odufisan, Osekhonmen Victory Abhulimen, Erastus Olarenwaju Ogunti. "Harnessing artificial intelligence and machine learning for fraud detection and prevention in Nigeria." *Journal of Economic Criminology*, Volume 7, 2025, Article 100127. ISSN 2949-7914. DOI: <https://doi.org/10.1016/j.jeconc.2025.100127>.

CONFERENCES

- 2025 Annual Conference of School of Electrical Systems Engineering (SESE) and School of Infrastructure, Minerals and Manufacturing Engineering (SIMME), FUTA: Innovative Solutions for Sustainable Living and Environmental Challenges. 22nd – 24th July.
- 2024 Federal University of Technology, Akure 165th Inaugural Lecture: Machine Learning: Pattern Matching and Classification Experience with Technology to Solve Human Problems. 12th March.

RESEARCH EXPERIENCE

Research Assistant – Machine Learning Research Lab, FUTA. | [Link](#) (July 2024 - September 2024)

- Investigated the impact of fraud on Nigeria's digital economy across finance, e-commerce, healthcare, and education sectors.
- Examined various AI methodologies, including supervised, unsupervised, and deep learning, and their use in anomaly detection, behavioral analysis, risk scoring, and network analysis.
- Proposed the future direction for AI and ML in enhancing financial crime prevention in the Nigerian digital economy.

Research Assistant – Machine Learning Research Lab, FUTA. | [Link](#) (February 2024 - March 2024)

- Tackled rising cyberattacks on Nigerian universities, targeting unauthorized access, phishing, DDoS, malware, and data breaches.
- Built a machine learning-based IDPS for real-time threat detection and prevention.
- Developed a modular pipeline with preprocessing, VAEs, one-hot encoding, and ensemble models; deployed via FastAPI; tested scalability with Apache JMeter; evaluated model drift.
- Delivered a scalable IDPS with accurate real-time threat detection, enhancing university cybersecurity.

Research Assistant – Machine Learning Research Lab, FUTA. | [Link](#) (January 2021 – August 2021)

- Supported research on Cross-Racial Automatic Age Estimation from Facial Images using Deep Learning by implementing transfer learning with ResNet50 and optimizing model performance through parameter tuning.
- Collaborated with researchers to prepare bi-weekly progress reports and provided administrative support to the team.
- Contributed to the lab's efforts in advancing machine learning applications for facial image analysis.

WORK EXPERIENCE

Full Stack Software Engineer – NEG AI Solutions (December 2024 – Present).

- Engineered an Intelligent Payment Routing System using Random Forest and Thompson Sampling Contextual Bandits for optimized transaction pathways, integrated with Flutterwave's API and a Gradio GUI for dynamic routing decisions. | [GitHub](#)
- Created a Predictive Analytics System for Payment Failures & Chargebacks, utilizing RNNs (LSTMs) and transformer models for sequence modeling of transaction data, delivering insights via a real-time analytics dashboard. | [GitHub](#)
- Developed Lejar: Personal Finance App, a fintech solution simplifying personal finance for Nigerian users with AI-powered receipt scanning and Naira-based budgeting, built using Next.js for secure, scalable financial management. | [GitHub](#)
- Created Crypto Asset Tracking & Portfolio Management App (Cointomic), a cross-platform fintech app for cryptocurrency trading with real-time data, secure authentication via Clerk, and Drizzle ORM, built using React Native, Expo 53, and CoinMarketCap APIs. | [GitHub](#)
- Designed CarbuCarbu Ride-Sharing App from scratch using React Native, Expo, and Clerk, integrating Google OAuth, email/password authentication, Google Maps for real-time ride tracking, Stripe payments, and a responsive UI with Tailwind CSS. | [GitHub](#)
- Built a Banking Application, a scalable digital platform for secure wallet management, KYC verification, and transactions in Nigeria, using Node.js, Express, MongoDB, Flutterwave for payments, SmileID for KYC, and Next.js, React, Tailwind CSS with real-time notifications via Socket.io, ensuring CBN and NDPC compliance. | [GitHub](#)
- Developed an AI-Powered Task Management System, a full-stack platform for task management and collaboration, integrating Pinata for file storage, JWT authentication, WebSockets for real-time chat, and performance analytics, built with Node.js, Express, MongoDB, React, Vite, and Tailwind CSS, validated via Jest tests. | [GitHub](#)

Software Engineering Instructor – Aptech Nigeria (NYSC) (October 2024 – May 2025).

- Teaching software engineering principles from basic to advanced, covering RESTful APIs and microservices architecture.
- Guiding students in real-world software development, including secure coding practice and API security.

Machine Learning & Software Engineer Intern - Chronicles Software Development Company (January 2023 – June 2024).

- Developed a student grade prediction model to forecast performance in Lagos state's West African Senior School Certificate Examination.
 - Built a learning platform and chatbot tutor using Natural Language Processing to assist students.
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RELEVANT COURSEWORKS

Artificial Neural Network, Robotics and Automation, Operating System, Computer Organization and Architecture, Data Communication and Network, Telecommunications Engineering, Digital Systems Design, Embedded Systems Design, Computer Graphics and Animation, Advanced Engineering Mathematics, Software Engineering, Software Development Techniques, Statistics, Programming in Python, C, and C#.

PROFESSIONAL DEVELOPMENT & CERTIFICATIONS

- Foundations of Cybersecurity (Google) | [CERTIFICATE](#) (December 2023).
 - ML Engineering for Production Specialization (DeepLearning.AI) | [CERTIFICATE](#) (September 2022).
 - Introduction to Machine Learning in Production (DeepLearning.AI) | [CERTIFICATE](#) (September 2022).
 - Machine Learning Data lifecycle in Production (DeepLearning.AI) | [CERTIFICATE](#) (September 2022).
 - Machine Learning Modelling Pipelines in Production (DeepLearning.AI) | [CERTIFICATE](#) (September 2022).
 - Deploying Machine Learning Models in Production (DeepLearning.AI) | [CERTIFICATE](#) (September 2022).
 - Deep Learning Specialization (DeepLearning.AI) | [CERTIFICATE](#) (September 2022).
 - Neural Networks and Deep Learning (DeepLearning.AI) | [CERTIFICATE](#) (September 2022).
 - Improving DNN: Hyperparameter Tuning, Regularization and Optimization (DL.AI) | [CERTIFICATE](#) (September 2022).
 - Structuring Machine Learning Projects (DeepLearning.AI) | [CERTIFICATE](#) (September 2022).
 - Convolutional Neural Network (DeepLearning.AI) | [CERTIFICATE](#) (September 2022).
 - Sequence Models (DeepLearning.AI) | [CERTIFICATE](#) (September 2022).
 - Machine Learning Specialization (DeepLearning.AI) | [CERTIFICATE](#) (July 2022).
 - Supervised Machine Learning: Regression and Classification (Stanford) | [CERTIFICATE](#) (July 2022).
 - Unsupervised Learning, Recommenders, Reinforcement Learning (Stanford) | [CERTIFICATE](#) (July 2022).
 - Advanced Learning Algorithms (Stanford) | [CERTIFICATE](#) (July 2022).
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SKILLS

- **Machine Learning & Data Science:** Python, TensorFlow, PyTorch, Scikit-learn, Pandas, NumPy, OpenCV
 - **Full Stack Development:** JavaScript, React, Next.js, React Native, Node.js, Express.js, Java, Spring, Hibernate, REST, GraphQL, OAuth 2.0, JWT, SSL/TLS, TailwindCSS
 - **Database Management:** MySQL, MongoDB, Redis, SQL
 - **Version Control & Testing:** Git, GitHub, JUnit, Jest, Supertest, Playwright
 - **Research & Analytical Skills:** Project Reports, Information analysis, synthesis, problem-solving
 - **Soft Skills:** Teamwork, communication, time management
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LEADERSHIP EXPERIENCE

CDS President for ICT, National Youth Service Corps, Ojodu LG (October 2024 – February 2025).

- Led the Information and Communication Technology group, organizing and managing projects, workshops, and training sessions to enhance digital literacy and develop technology-driven solutions for local community challenges.
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ENGLISH PROFICIENCY

Test Report Form Number: 23NG037884ODUO1500A - International English Language Testing System (November 2023).

Overall Band Score: 7.5 & CEFR Level: C1 (Advanced English Language User)

INTERESTS: Travelling, Observation and Research, Music, Sport, Reading, Leadership, and Entrepreneurship Workshops.

REFERENCE: Available on request